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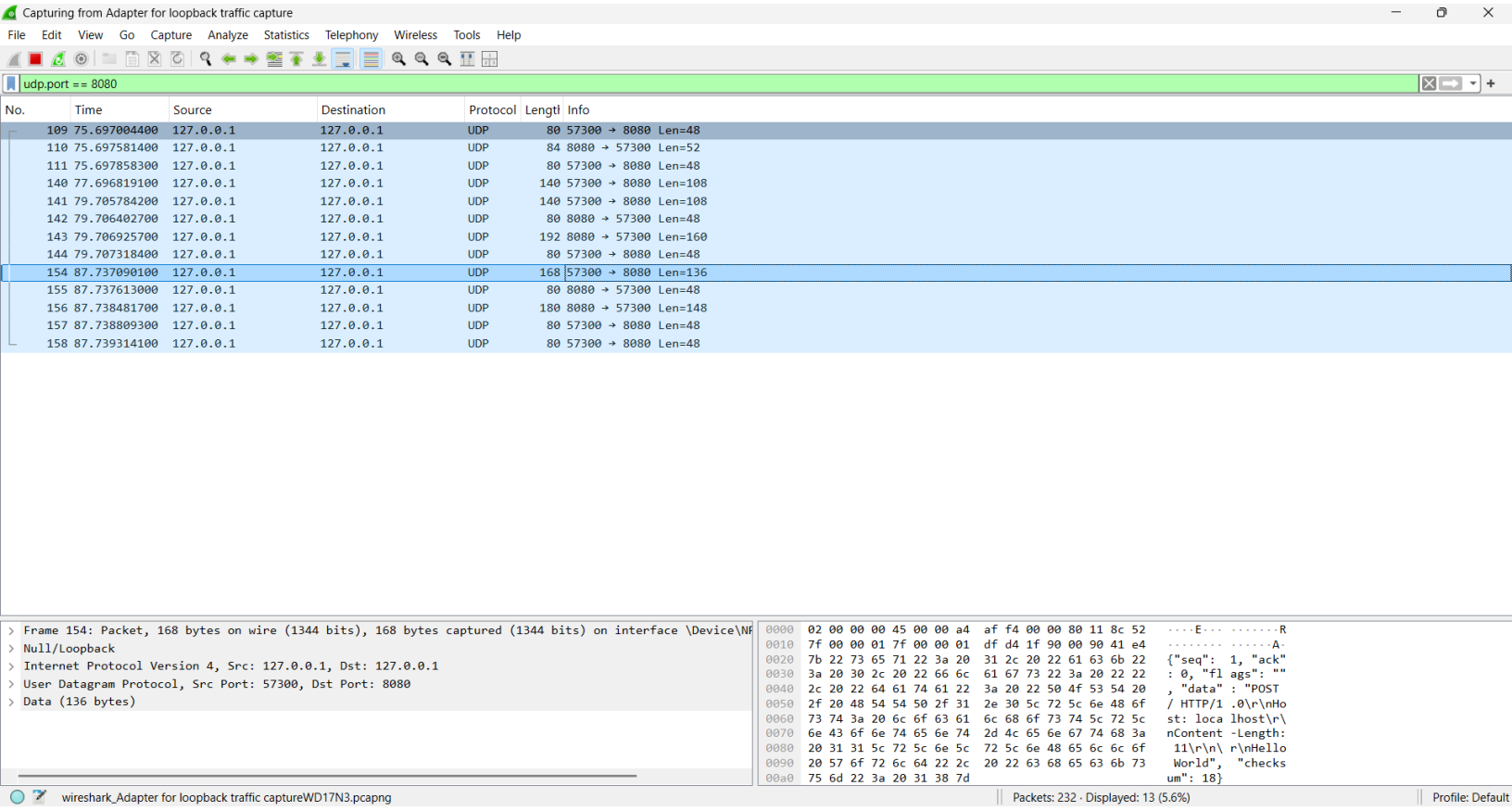
# Lab 4

## - Wireshark connections

1) Connection when running client and server http using terminal



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## 2) Connection http server to web browser using proxy

Capturing from Adapter for loopback traffic capture

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tcp.port == 8000

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 393 | 262.450853200 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 52869 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 Information _PERM |
| 394 | 262.451097800 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 8000 → 52869 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 395 | 262.451176400 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 52869 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |
| 396 | 262.469966100 | 127.0.0.1 | 127.0.0.1 | HTTP | 727 | GET /hello HTTP/1.1 |
| 397 | 262.470036600 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 8000 → 52869 [ACK] Seq=1 Ack=684 Win=64768 Len=0 |
| 399 | 262.486600700 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 51529 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 400 | 262.486768700 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 8000 → 51529 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 401 | 262.486848800 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 51529 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |
| 428 | 283.176026400 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 52869 → 8000 [FIN, ACK] Seq=684 Ack=1 Win=65280 Len=0 |
| 429 | 283.176089200 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 8000 → 52869 [ACK] Seq=1 Ack=685 Win=64768 Len=0 |
| 444 | 297.959090500 | ::1 | ::1 | TCP | 76 | 64335 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM |
| 445 | 297.959122700 | ::1 | ::1 | TCP | 64 | 8000 → 64335 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 |
| 446 | 298.004611900 | ::1 | ::1 | TCP | 76 | 53839 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM |
| 447 | 298.004639900 | ::1 | ::1 | TCP | 64 | 8000 → 53839 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 |
| 448 | 298.222985400 | ::1 | ::1 | TCP | 76 | 57991 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM |
| 449 | 298.223024600 | ::1 | ::1 | TCP | 64 | 8000 → 57991 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 |
| 450 | 298.273429200 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 60865 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 451 | 298.273637100 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 8000 → 60865 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 452 | 298.273723600 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 60865 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |
| 453 | 298.274622400 | 127.0.0.1 | 127.0.0.1 | HTTP | 722 | GET / HTTP/1.1 |
| 454 | 298.274686700 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 8000 → 60865 [ACK] Seq=1 Ack=679 Win=64768 Len=0 |
| 455 | 298.306327700 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 56448 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 456 | 298.306524500 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 8000 → 56448 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 457 | 298.306602700 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 56448 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |
| 458 | 298.537576400 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 51310 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 459 | 298.537793100 | 127.0.0.1 | 127.0.0.1 | TCP | 56 | 8000 → 51310 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM |
| 460 | 298.537875100 | 127.0.0.1 | 127.0.0.1 | TCP | 44 | 51310 → 8000 [ACK] Seq=1 Ack=1 Win=65280 Len=0 |

```
> Frame 396: Packet, 727 bytes on wire (5816 bits), 727 bytes captured (5816 bits) on interface \Device\NF
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 52869, Dst Port: 8000, Seq: 1, Ack: 1, Len: 683
> Hypertext Transfer Protocol
```

```
0000  02 00 00 00 45 00 02 d3  b0 dc 40 00 80 06 49 46   ····E··· ··@···IF
0010  7f 00 00 01 7f 00 00 01  ce 85 1f 40 6b 92 69 28   ........ ···@k·i(
0020  58 f3 44 d0 50 18 00 ff  23 23 00 00 47 45 54 20   X·D·P··· ##··GET
0030  2f 68 65 6c 6c 6f 20 48  54 54 50 2f 31 2e 31 0d   /hello H TTP/1.1·
0040  0a 48 6f 73 74 3a 20 31  32 37 2e 30 2e 30 2e 31   ·Host: 1 27.0.0.1
0050  3a 38 30 30 30 0d 0a 43  6f 6e 6e 65 63 74 69 6f   :8000··C onnectio
0060  6e 3a 20 6b 65 65 70 2d  61 6c 69 76 65 0d 0a 73   n: keep- alive··s
0070  65 63 2d 63 68 2d 75 61  3a 20 22 4d 69 63 72 6f   ec-ch-ua : "Micro
0080  73 6f 66 74 20 45 64 67  65 22 3b 76 3d 22 31 34   soft Edg e";v="14
0090  37 22 2c 20 22 4e 6f 74  2e 41 2f 42 72 61 6e 64   7", "Not .A/Brand
00a0  22 3b 76 3d 22 38 22 2c  20 22 43 68 72 6f 6d 69   ";v="8", "Chromi
```

Capturing from Adapter for loopback traffic capture

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udp.port == 8080

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 109 | 75.697004400 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57300 → 8080 Len=48 |
| 110 | 75.697581400 | 127.0.0.1 | 127.0.0.1 | UDP | 84 | 8080 → 57300 Len=52 |
| 111 | 75.697858300 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57300 → 8080 Len=48 |
| 140 | 77.696819100 | 127.0.0.1 | 127.0.0.1 | UDP | 140 | 57300 → 8080 Len=108 |
| 141 | 79.705784200 | 127.0.0.1 | 127.0.0.1 | UDP | 140 | 57300 → 8080 Len=108 |
| 142 | 79.706402700 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 8080 → 57300 Len=48 |
| 143 | 79.706925700 | 127.0.0.1 | 127.0.0.1 | UDP | 192 | 8080 → 57300 Len=160 |
| 144 | 79.707318400 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57300 → 8080 Len=48 |
| 154 | 87.737090100 | 127.0.0.1 | 127.0.0.1 | UDP | 168 | 57300 → 8080 Len=136 |
| 155 | 87.737613000 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 8080 → 57300 Len=48 |
| 156 | 87.738481700 | 127.0.0.1 | 127.0.0.1 | UDP | 180 | 8080 → 57300 Len=148 |
| 157 | 87.738809300 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57300 → 8080 Len=48 |
| 158 | 87.739314100 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57300 → 8080 Len=48 |
| 398 | 262.471099900 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 49794 → 8080 Len=48 |
| 1962 | 485.782462300 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 51483 → 8080 Len=48 |
| 1963 | 485.782823300 | 127.0.0.1 | 127.0.0.1 | UDP | 84 | 8080 → 51483 Len=52 |
| 1964 | 485.782983800 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 51483 → 8080 Len=48 |
| 1965 | 485.792245600 | 127.0.0.1 | 127.0.0.1 | UDP | 818 | 51483 → 8080 Len=786 |
| 1967 | 487.795329900 | 127.0.0.1 | 127.0.0.1 | UDP | 818 | 51483 → 8080 Len=786 |
| 1968 | 487.796097000 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 8080 → 51483 Len=48 |
| 1969 | 487.797103500 | 127.0.0.1 | 127.0.0.1 | UDP | 175 | 8080 → 51483 Len=143 |
| 1970 | 487.797397800 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 51483 → 8080 Len=48 |
| 1975 | 487.797685900 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 51483 → 8080 Len=48 |
| 1980 | 487.837659300 | 127.0.0.1 | 127.0.0.1 | UDP | 80 | 57475 → 8080 Len=48 |

```
> Frame 1969: Packet, 175 bytes on wire (1400 bits), 175 bytes captured (1400 bits) on interface \Device\N
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 8080, Dst Port: 51483
> Data (143 bytes)
```

```
0000  02 00 00 00 45 00 00 ab  b6 e7 00 00 80 11 85 58   ····E··· ·******X
0010  7f 00 00 01 7f 00 00 01  1f 90 c9 1b 00 97 d1 00   ........ ********
0020  7b 22 73 65 71 22 3a 20  30 2c 20 22 61 63 6b 22   {"seq":  0, "ack"
0030  3a 20 30 2c 20 22 66 6c  61 67 73 22 3a 20 22 22   : 0, "fl ags": ""
0040  2c 20 22 64 61 74 61 22  3a 20 22 48 54 54 50 2f   , "data" : "HTTP/
0050  31 2e 30 20 32 30 30 20  4f 4b 5c 72 5c 6e 43 6f   1.0 200  OK\r\nCo
0060  6e 74 65 6e 74 2d 54 79  70 65 3a 20 74 65 78 74   ntent-Ty pe: text
0070  2f 70 6c 61 69 6e 5c 72  5c 6e 43 6f 6e 74 65 6e   /plain\r \nConten
0080  74 2d 4c 65 6e 67 74 68  3a 20 39 5c 72 5c 6e 5c   t-Length : 9\r\n\
0090  72 5c 6e 48 6f 6d 65 20  50 61 67 65 22 2c 20 22   r\nHome  Page", "
00a0  63 68 65 63 6b 73 75 6d  22 3a 20 32 34 35 7d      checksum ": 245}
```

wireshark Adapter for loopback traffic captureWD17N3.pcapng | Packets: 2167 · Displayed: 24 (1.1%) | Profile: Default

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# Test Cases:

PS D:\Documents\Spring 26\Networks\Lab\Lab4\-simulation-for-TCP-packets-using-a-UDP-connection> python test_cases.py

Test 1: handshake

Connected!

Client: Handshake OK

Client connected

Connection closed

Server: Handshake OK

Connection closed

Test 2: retransmission

Connected!

Client connected

Sent: {"seq": 0, "ack": 0, "flags": "", "data": "Test Message with Loss", "checksum": 34}

Received: {'seq': 0, 'ack': 0, 'flags': '', 'data': 'Test Message with Loss', 'checksum': 34}

Server received: Test Message with Loss

ACK received

Connection closed

Connection closed

Test 3: duplicate handling

Connected!

Client connected

Sent: {"seq": 0, "ack": 0, "flags": "", "data": "Duplicate Test", "checksum": 91}

Received: {'seq': 0, 'ack': 0, 'flags': '', 'data': 'Duplicate Test', 'checksum': 91}

Server received first: Duplicate Test

ACK received

Sent: {"seq": 1, "ack": 0, "flags": "", "data": "Duplicate Test", "checksum": 91}

Packet lost

Timeout → Resending...

Timeout → Resending...

Sent: {"seq": 1, "ack": 0, "flags": "", "data": "Duplicate Test", "checksum": 91}

Received: {'seq': 1, 'ack': 0, 'flags': '', 'data': 'Duplicate Test', 'checksum': 91}

Server received second: Duplicate Test

ACK received

Connection closed

Connection closed

All tests completed!